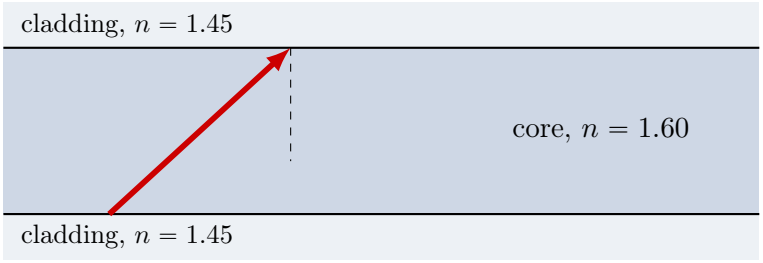
1. Green light with a vacuum wavelength of 520 nm enters a diamond ($n = 2.42$).

$\lambda_n =$ _____ $f_n =$ _____ $v_n =$ _____

A. Find the wavelength, frequency, and speed of the light **inside** the diamond. Show your work below and fill in the blanks.

B. Which of the three did **not** change when the light entered the diamond? Why not?

2. An optical fiber has a glass core ($n = 1.60$) wrapped in a cladding ($n = 1.45$).



A. What is the critical angle for light inside the core hitting the cladding?

B. The ray in the figure hits the wall at 48° from the normal. Does it stay in the core? On the figure, draw what happens next.

C. Could light going from the **cladding** into the **core** ever have total internal reflection? Why or why not?

3. A narrow beam of white light hits a block of flint glass at 50° from the normal. For this glass, $n_{\text{red}} = 1.61$ and $n_{\text{violet}} = 1.65$.

A. Find the refracted angle for red light and for violet light. What is the angle between them inside the glass?

B. Which color bends more? Explain in terms of how fast each color travels in the glass.

C. Sketch the white beam entering the glass and splitting. Label the red and violet rays and the normal.